

MATH 3260 - Chapter 5 Worksheets

Inner Product Spaces

Name: _____

Date: _____

These are student practice worksheets. They are designed to be used with the chapter notebook and notes, but every problem below contains the data needed to work it independently. No solutions are included in this student version.

Section 5.1 - Length and Dot Product in $\mathbb{R}^n$

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. Find the norm of $(6, -3, 6)$ and a unit vector in the same direction. Verify explicitly that your unit vector has norm 1.
2. Find the distance between $(1, 4, -2)$ and $(5, -2, 1)$.
3. Compute the dot product and the angle between $(1, 2, 2)$ and $(2, 0, -1)$.
4. Determine whether $(2, -1, 4)$ and $(1, 6, 1)$ are orthogonal. State your conclusion using the symbol $\perp$ when appropriate.
5. Verify the Cauchy-Schwarz inequality for $(2, 1)$ and $(-1, 3)$.
6. Write the dot product of (a, b, c) and (x, y, z) as a matrix product.

Section 5.2 - Inner Product Spaces

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. For

$$\langle \mathbf{u}, \mathbf{v} \rangle = 3u_1v_1 + 2u_2v_2,$$

find $\langle (2, -1), (4, 3) \rangle$ and the induced norm of $(2, -1)$.

2. Verify all four real inner-product axioms for

$$\langle \mathbf{u}, \mathbf{v} \rangle = 3u_1v_1 + 2u_2v_2.$$

State clearly where positivity of the weights is used.

3. Explain why

$$u_1v_1 + u_2v_2 - u_3v_3$$

is not an inner product on $\mathbb{R}^3$. It is enough to identify one failed axiom with a counterexample.

4. Using the coefficient inner product, find

$$\langle 1 + x + 2x^2, 2 - 3x + x^2 \rangle.$$

5. On $C[0, 1]$, compute

$$\langle 1, x^2 \rangle = \int_0^1 x^2 dx.$$

6. Find the projection of $(5, 1, 4)$ onto $(1, 0, 2)$.

7. Verify that the residual from Problem 6 is orthogonal to the projection direction.

Section 5.3 - Orthonormal Bases and Gram-Schmidt

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. Determine whether

$$\{(1, 1, 0), (1, -1, 0), (0, 0, 2)\}$$

is orthogonal. Show the pairwise dot products. Is the set orthonormal?

2. Normalize the set in Problem 1.

3. Find the coordinates of $(4, 2, 3)$ relative to the normalized basis from Problem 2. Verify by reconstruction.

4. Apply Gram-Schmidt to $(1, 0)$ and $(1, 2)$.

5. Apply Gram-Schmidt to $(1, 1, 0)$ and $(1, 0, 1)$.

6. Find an orthonormal basis for

$$\text{span}\{(1, 1, 1), (1, -1, 0)\}.$$

At each stage, identify which projection is being removed.

Section 5.4 - Orthogonal Complements, Fundamental Subspaces, and Least Squares

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. Find a basis for

$$\text{span}\{(1, 1, 0), (0, 1, 1)\}^\perp.$$

2. Project $(4, 5, 6)$ onto the line spanned by $(1, 1, 1)$.

3. Project $(3, 1, 4)$ onto the plane spanned by the orthogonal vectors $(1, 1, 0)$ and $(1, -1, 2)$.

4. For

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{bmatrix},$$

find:

- (a) $\text{rank}(A)$;
- (b) a basis for $\text{Row}(A)$;
- (c) a basis for $\text{Col}(A)$;
- (d) a basis for $\text{Null}(A)$;
- (e) a basis for the left nullspace $\text{Null}(A^T)$.

Then verify the two dimension relationships suggested by rank-nullity.

5. Find the least-squares line for

$$(0, 1), (1, 2), (2, 2), (3, 5).$$

6. Verify that the least-squares residual from Problem 5 is orthogonal to both columns of the design matrix.

Section 5.5 - Applications of Inner Product Spaces

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

Keep the operation types separate: a dot product returns a scalar, a cross product in $\mathbb{R}^3$ returns a vector, and scalar multiplication rescales a vector.

1. Find

$$(2, 1, -1) \times (1, -2, 3).$$

2. Verify that the answer to Problem 1 is orthogonal to both original vectors.
3. Find the parallelogram area and triangle area determined by the two vectors in Problem 1.
4. Find a unit normal to the plane spanned by $(1, 0, 1)$ and $(0, 2, 1)$.
5. Find the best constant approximation to $f(x) = x$ on $[0, 2]$. Show the integral equation that determines the best constant.
6. Write the second-order Fourier approximation of $f(x) = x - \pi$ on $[0, 2\pi]$ using the coefficient formulas from the notes.